Assignment 2

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Q.1 What kind of information should be sought in interviews?

Interviews are used in system analysis and requirement gathering to collect detailed, qualitative information from users & stakeholders. The following types of information should be sought :-

i) Organizational Information
- Organizational structure
- Roles and responsibilities
- Reporting relationships

ii) Current System Information
- How the existing system works
- Current procedures & workflows
- problems & limitations of the current system

iii) User requirements
- User needs & expectations ~~from the new system.~~
- ~~Data requirements, input, output~~ Input processing, and output requirements.
- frequency & volume of tasks

iv) Decision making Information
- How decisions are made
- Data needed for decision making
- Reports & summaries required

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iv) Problems & Pain points
- Errors, delays, redaundancies
- User complaints & difficulties

v) Suggestions & Improvements
- User ideas for system improvement
- Desired features in the new system.

Q. 2 Describe the commonly used methods for performing economic cost-benifit analysis.

→ Economic feasibility or cost-benefit analysis is the analysis of system if the expected benifit equal or exceed the expected cost, the main purpose is to identify the financial benefits and costs associated with the development project. During project initiation and planning, it will be impossible for us to define precisely all benifits and costs a related to particular projects. We review worksheets to record costs & benifits. These worksheets are used after each SDLC phase to decide wheather to continue, redirect or kill or project.

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- Common methods are :-
  - Payback period
  - Return on Investment (~~NPV~~ ROI)
  - Net present value (NPV)
  - Benefit cost Ratio (BCR)
  - Break - Even Analysis

Q. 3 Explain the steps in feasibility analysis.

→ feasibility analysis determines wheather a system is worth developing. The main steps are:

1. Define the problem
   - ~~Understand~~ system objectives.

2. Identify alternatives
   - Different possible solⁿ.

3. Technical feasibility
   - Availability of hardware, software skills.

4. Economic feasibility
   - cost vs benifit analysis

5. Operational feasibility
   - User acceptance & usability

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6. Schedule feasibility
- completion within time unit

7. Decision making
- select best feasible alternative

Q.4) Explain the payback period with an example.
→ Payback period is the time required to recover the initial investment. from ~~this~~ the systems net benefits.

$$\text{payback period} = \frac{\text{Initial Investment}}{\text{cash in flows}}$$

For eg:-
If a system cost Rs. 100000 and gives Rs. 25000 profit per year:

$$\text{payback period} = \frac{100,000}{25,000}$$
$$= 4 \text{ years.}$$

It is one of the simplest & most commonly used methods of economic feasibility analysis in system development. In system analysis & design, the payback period helps management decide how quickly the cost of proposed system will be recovered.

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Q.5.) Why feasibility analysis is necessary before designing a system?

→ Feasibility analysis determines whether a system is worth developings.

Feasibility analysis is necessary because:

i) It ensure the system is technically possible.
ii) It checks ecomic profitability
iii) It confirms user acceptance
iv) It reduces risk & failure
v) It save time & resources

Feasibility analysis is a critical step before system design because it ensures the system is technically possible, economically profitable, operationally acceptable & time feasible. without feasibility analysis system development is risky & may lead to failure.

Q.6.) A system cost Rs. 100000 to install & Rs. 8000 per month as recuring expenses. The benifit per year is 150000. Assuming an interest rate is 12% what is the payback period of the investment.

→ Soln Given.

installation cost = Rs. 100000
monthly recurring cost = 8000
Annual benifit = Rs. 150000
Interest Rate = 12%

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Annual recurring cost = 8000 × 12
= 96000

Annual net benefit = 150000 - 96000
= 54000

Payback period = $\frac{\text{initial investment}}{\text{cash in flow}}$

= $\frac{100000}{54000}$

= 1.85 years

3 Marks